

Bike Sharing Demand Analytics Project Documentation

1. Project Overview

The Bike Sharing Demand Analytics project is a Power BI-based data analysis project developed to understand bike rental patterns using historical bike-sharing data.

The analysis focuses on rental demand across different time periods, seasons, weather conditions, working days, and user types.

The project demonstrates the complete Power BI workflow:

1. Dataset collection
2. Data cleaning and transformation
3. Data modeling
4. DAX measure creation
5. Data analysis
6. Dashboard development
7. Interactive filtering

2. Project Objective

The primary objective of this project is:

> To analyze bike rental demand across time, user types, weather conditions, seasons, and working days in order to identify demand patterns and factors affecting bike utilization.

The dashboard is designed to answer the following questions:

- How many bikes are rented overall?
- How does demand change over time?
- Which hours experience the highest demand?
- How does demand vary by season?

- How does weather influence rental demand?
- What proportion of rentals comes from registered and casual users?
- How does rental demand compare between years?
- How does cumulative rental demand develop throughout the year?

3. Dataset

The project uses the **UCI Bike Sharing Dataset**.

The dataset contains bike rental information collected from the Capital Bikeshare system for the years 2011 and 2012.

The hourly dataset contains 17,389 observations.

Dataset Source

UCI Machine Learning Repository

<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

DOI:

10.24432/C5W894

The dataset is provided under the CC BY 4.0 license.

4. Raw Dataset

The original dataset is stored in the:

Raw Dataset/

folder.

The primary file used for analysis is:

`hour.csv`

The original ZIP archive is also retained for reference.

Important Columns

Column	Description
dteday	Date
hr	Hour
season	Season category
yr	Year
mnth	Month
holiday	Whether the day is a holiday
weekday	Day of the week
workingday	Whether it is a working day
weathersit	Weather condition
temp	Normalized temperature
atemp	Normalized feeling temperature
hum	Humidity
windspeed	Wind speed
casual	Casual user rentals
registered	Registered user rentals
cnt	Total bike rentals

5. Data Cleaning and Transformation

The dataset was imported into Power BI and transformed using Power Query.

The main transformations included:

- Renaming columns for readability
- Assigning appropriate data types
- Converting the date column to Date format
- Creating meaningful category names for coded values
- Preparing fields for analysis and visualization
- Creating a proper calendar/date table for time intelligence

Examples of categorical transformations include:

Season

Original Value	Meaning
1	Winter
2	Spring
3	Summer
4	Fall

Working Day

Original Value	Meaning
0	Non-Working Day
1	Working Day

Year

Original Value	Meaning
0	2011
1	2012

6. Data Modeling

A structured data model was created to support efficient analysis and time-based calculations.

The main fact table contains the bike rental observations.

A separate calendar/date dimension was created to support time intelligence calculations.

Main Tables

Fact Bike Rentals

Contains the rental transactions/observations and measures such as:

- Total Rentals
- Registered Rentals
- Casual Rentals
- Hour
- Weather
- Season
- Working Day
- Temperature
- Humidity
- Wind Speed

Calendar Dimension

The calendar table contains fields such as:

- Date
- Year
- Month
- Month Number
- Quarter
- Day

- Day Name
- Weekday Number

The relationship is established between the Date field in the calendar table and the Date field in the fact table.

The conceptual and physical data models are available in:

Data models/

7. DAX Measures

Total Rentals

```
Total Rentals =  
SUM(FactBikeRentals[Total Rentals])
```

Calculates the total number of bike rentals.

Registered Rentals

```
Registered Rentals =  
SUM(FactBikeRentals[Registered Rentals])
```

Calculates the total rentals made by registered users.

Casual Rentals

```
Casual Rentals =  
SUM(FactBikeRentals[Casual Rentals])
```

Calculates the total rentals made by casual users.

Average Daily Rentals

```
Average Daily Rentals =  
AVERAGEX(  
    VALUES(DateTable[Date]),  
    [Total Rentals]  
)
```

This measure uses `AVERAGEX` to iterate through each distinct date and calculate the average daily rental count.

This demonstrates the use of an iterator rather than simply calculating the average of individual hourly records.

Previous Year Rentals

```
Previous Year Rentals =  
CALCULATE(  
    [Total Rentals],  
    SAMEPERIODLASTYEAR(DateTable[Date])  
)
```

Calculates rental demand for the corresponding period in the previous year.

YoY Growth %

```
YoY Growth % =  
DIVIDE(  
    [Total Rentals] - [Previous Year Rentals],  
    [Previous Year Rentals]  
)
```

Calculates the percentage change in rentals compared with the previous year.

YTD Rentals

```
YTD Rentals =  
TOTALYTD(  
    [Total Rentals],  
    DateTable[Date]  
)
```

Calculates cumulative rental demand from the beginning of the year up to the current date.

8. Evaluation Context

The project demonstrates how Power BI evaluates measures based on the current filter context.

For example, when the user selects a particular season using the slicer, the same `[Total Rentals]` measure automatically recalculates based on the selected season.

Similarly, selecting a particular month, weather condition, or working-day category changes the filter context and updates the dashboard visuals.

9. Filter Manipulation

Filter manipulation is demonstrated primarily through:

- Power BI slicers
- `CALCULATE`
- `SAMEPERIODLASTYEAR`
- Time-intelligence functions

For example:

```
Previous Year Rentals =  
CALCULATE(  
    [Total Rentals],  
    SAMEPERIODLASTYEAR(DateTable[Date])  
)
```

`CALCULATE` modifies the existing filter context to evaluate rental demand for the previous-year period.

10. Dashboard Components

The dashboard contains five KPI cards.

KPI 1 – Total Rentals

Shows the total number of bike rentals.

KPI 2 – Registered Rentals

Shows rentals made by registered users.

KPI 3 – Casual Rentals

Shows rentals made by casual users.

KPI 4 – Average Daily Rentals

Shows the average number of rentals per day.

KPI 5 – YoY Growth %

Shows the percentage change in rental demand compared with the previous year.

11. Dashboard Visualizations

Rental Demand Trend

Shows how total rental demand changes over time.

This helps identify overall growth and seasonal patterns.

Rental Demand by Hour

Shows rental demand across different hours of the day.

This helps identify peak and low-demand periods.

Monthly Rentals: Current vs Previous Year

Compares monthly rental demand between the current and previous year.

This visual demonstrates the use of time intelligence and `SAMEPERIODLASTYEAR`.

Rental Demand by Season

Shows how bike rental demand varies across different seasons.

Rental Demand by Weather

Shows the relationship between weather conditions and bike rental demand.

Cumulative YTD Rentals

Shows cumulative rental demand throughout the year.

This visual demonstrates the use of the `TOTALYTD` time-intelligence function.

12. Interactive Filters

The dashboard provides the following slicers:

- Month
- Season
- Weather
- Working Day

These slicers allow users to dynamically explore different segments of the dataset.

13. Dashboard Design

A dark urban mobility theme was used for the dashboard.

The design uses:

- Dark navy background
- Blue/cyan primary data elements
- Teal highlights
- White primary text
- Muted secondary text
- Dark blue chart containers
- Consistent borders and spacing

The design was chosen to provide a modern analytical dashboard while maintaining readability.

14. Key Analytical Areas

The dashboard allows analysis of:

Time

Rental demand can be analyzed by:

- Year
- Quarter
- Month
- Day
- Hour

User Type

Rental demand can be separated into:

- Registered users
- Casual users

Weather

Rental demand can be compared across different weather conditions.

Season

Rental demand can be analyzed across:

- Winter
- Spring
- Summer
- Fall

Working Day

Rental patterns can be compared between:

- Working days
- Non-working days

15. Power BI Concepts Demonstrated

This project demonstrates the following Power BI concepts:

- Power Query data transformation
- Data types
- Data modeling
- Relationships
- Fact and dimension tables
- Star-schema concepts
- Calculated columns

- Measures
 - Aggregation functions
 - Iterator functions
 - Filter context
 - Evaluation context
 - CALCULATE
 - DIVIDE
 - Time intelligence
 - SAMEPERIODLASTYEAR
 - TOTALYTD
 - Interactive slicers
 - Dashboard design
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16. Repository Structure

```
1.Bike-Sharing-Demand-Analytics/  
├── Raw Dataset/  
│   ├── hour.csv  
│   ├── bike+sharing+dataset.zip  
│   └── bike+sharing+dataset/  
├── Dashboard Preview/  
│   └── dashboard_preview.png  
├── Data models/  
│   ├── 1. conceptual_model.png  
│   └── 2. Physical_model.png  
├── Power BI/  
│   └── Bike_Sharing_Demand_Analytics.pbix  
├── Documentation.pdf  
└── README.md
```

17. How to Use the Project

1. Clone or download this repository.
 2. Open the Power BI file located inside the `Power BI` folder.
 3. If required, update the dataset path in Power Query.
 4. Refresh the data.
 5. Explore the dashboard using the available slicers and visuals.
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18. Conclusion

This project demonstrates the complete process of building an analytical dashboard in Power BI, starting from a raw dataset and progressing through data transformation, modeling, DAX calculations, time intelligence, and visualization.

The dashboard provides an interactive way to explore bike-sharing demand and understand how rental patterns vary across time, weather, seasons, user types, and working days.